Harry Nguyen

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Education

Georgia State University **Atlanta, GA**

Bachelor of Science in Computer Science, GPA: 3.75/4.0 *Expected May 2027*

- **Coursework:** Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures

Work Experience

TiMoto AI **Sep 2025 – Present**

Software Engineer

- Prevented single-zone failures from taking production down, sustaining **99.9% uptime** and cutting infrastructure cost **44%** ($40-60/month), by architecting a **multi-AZ ECS Fargate** deployment with **Terraform IaC**, CloudWatch observability, and an auto-rollback circuit breaker pattern
- Restored **100% evaluation success rate** at **sub-50ms p99** by resolving a production **gRPC** deadlock – replaced inconsistent lock-acquisition ordering under concurrent calls with a single enforced sequencing rule across the inter-service layer
- Cut incident root-cause time from unknown to **under 15 minutes** by building TiMoto's observability stack from zero – deployed **Prometheus** and **Grafana** metrics alongside CloudWatch dashboards across the gRPC and ML serving layers
- Eliminated **OOM failures** under concurrent production load by replacing naive batching with a **vLLM/PagedAttention** inference engine, solving KV cache memory fragmentation for LLM serving

Google **May 2025 – Aug 2025**

Software Engineer Intern – Chrome Browser

- Shipped a **C++ IPC transport layer** to Chrome's stable channel – serving **3B+ active users** at **sub-50ms p99** and **10K+ req/sec** – by choosing Protocol Buffers over custom serialization for schema evolution and cross-language compatibility
- Turned a visibly sluggish settings search into an instant one for power users – **96%** p99 latency cut (1,200ms down to sub-50ms), zero production regressions – by replacing a mutex-contended search path with a **lock-free concurrent trie**
- Landed production-ready changes into Chrome's codebase – reviewed and adopted by senior engineers – by delivering design docs and code reviews at **95% test coverage** alongside the Chrome infrastructure team

Develop for Good **May 2024 – Aug 2024**

Software Engineer Intern

- Cut API response time from 3+ seconds to **sub-100ms** on **10,000+ record** datasets by diagnosing an N+1 query bottleneck and redesigning the data layer with **PostgreSQL indexing**
- Scaled a backend to support **500+ concurrent users** and cut deployment time **90%** by designing a stateless AWS BaaS with JWT auth and a GitHub Actions CI/CD auto-scaling policy

Personal Projects

Pulumi – Contributor | *Go, TypeScript, Infrastructure-as-Code (IaC)*

- Submitted Go CLI features and bug fixes enabling multi-cloud (AWS/Azure/GCP) provisioning – under active review by core maintainers
- Analyzed **Raft/Paxos consensus** in a distributed state synchronization layer – studied linearizability guarantees and correctness under concurrent operations and partial failures

Technical Skills

Distributed Systems: gRPC, Protocol Buffers, Circuit breakers, Exactly-once semantics, Fault tolerance, Kafka, Multi-AZ, Event-driven architecture
Cloud & Infrastructure: AWS (ECS Fargate, EKS, EC2, S3, RDS), Terraform, Kubernetes, Docker, CI/CD, CloudWatch, Prometheus, Grafana, GCP
Languages: C++, Go, Python, TypeScript, Java, JavaScript, SQL, Bash, Rust
Frameworks & Databases: Django, FastAPI, Node.js, React, REST APIs, PostgreSQL, Redis, MongoDB
ML & AI Infrastructure: vLLM, PagedAttention, Continuous Batching, LoRA/QLoRA, PyTorch, LangChain, LLM-as-a-judge evaluation
AI Dev Tools: Claude Code, GitHub Copilot, Codex, Cursor